

Lecture 6

ChE 467/596-012 Polymer Rheology

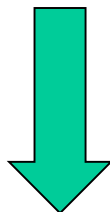
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Recap- Possible experiments

- **Steady shear**
- **Dynamic oscillatory shear**
- **Start-up of steady shear – stress growth**
- **Stress relaxation following steady shear**
- **Step strain -stress relaxation following sudden strain**
- **Creep and recovery**

Shear experiments

- Which experiment do we do?
- Can same or complementary information be obtained from two experiment?
- What molecular information do each experiment provide?

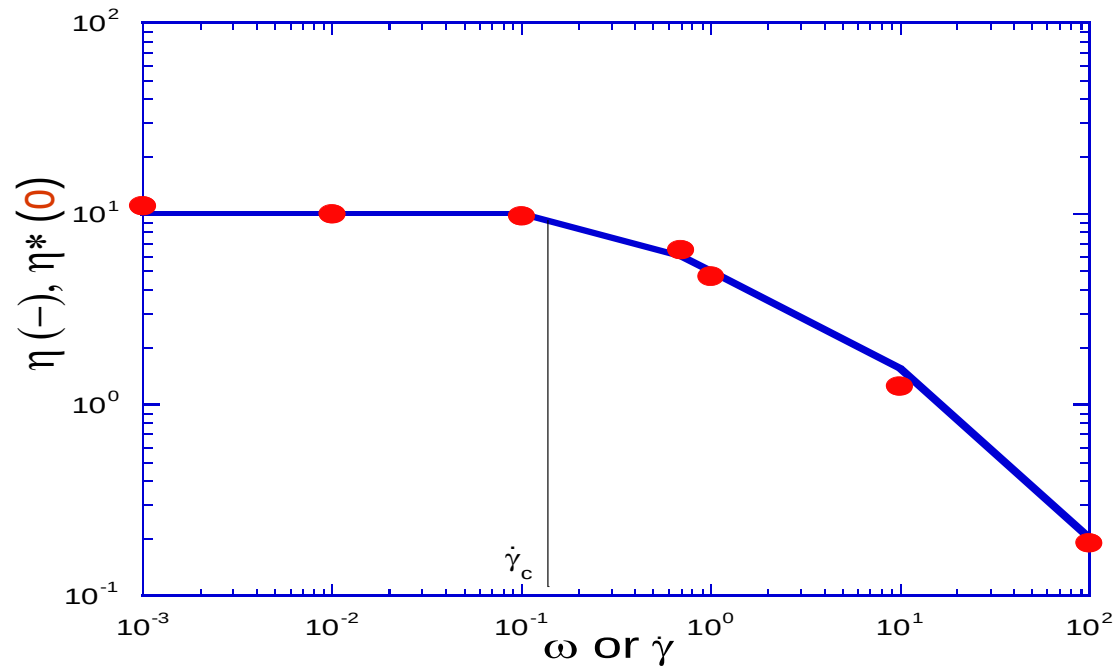


Correlation between Material functions

Effects of molecular properties on Material functions

Correlation Between Material Functions

Cox-Merz Rule



$$\eta^*(\omega) = \eta(\dot{\gamma}) \quad @ \quad \omega = \dot{\gamma}$$

$$\omega_c = \dot{\gamma}_c; \quad \frac{1}{\omega_c} \sim \frac{1}{\dot{\gamma}_c} = \lambda \quad \lambda \equiv \text{longest relaxation time}$$

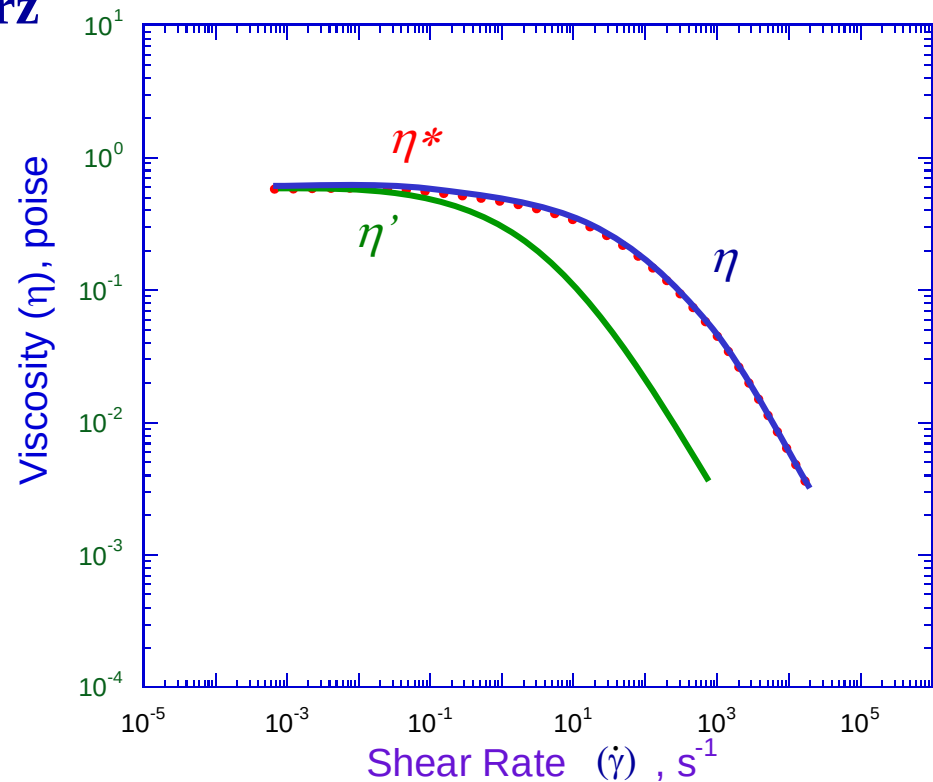
Dynamic vs Steady shear

Complex (η^*) vs steady (η) vs dynamic (η') viscosity

Already know from Cox-Merz

$$\eta^*(\omega) = \eta(\dot{\gamma}) \quad @ \quad \omega = \dot{\gamma}$$

$\eta = \eta'$ only at low shear rates



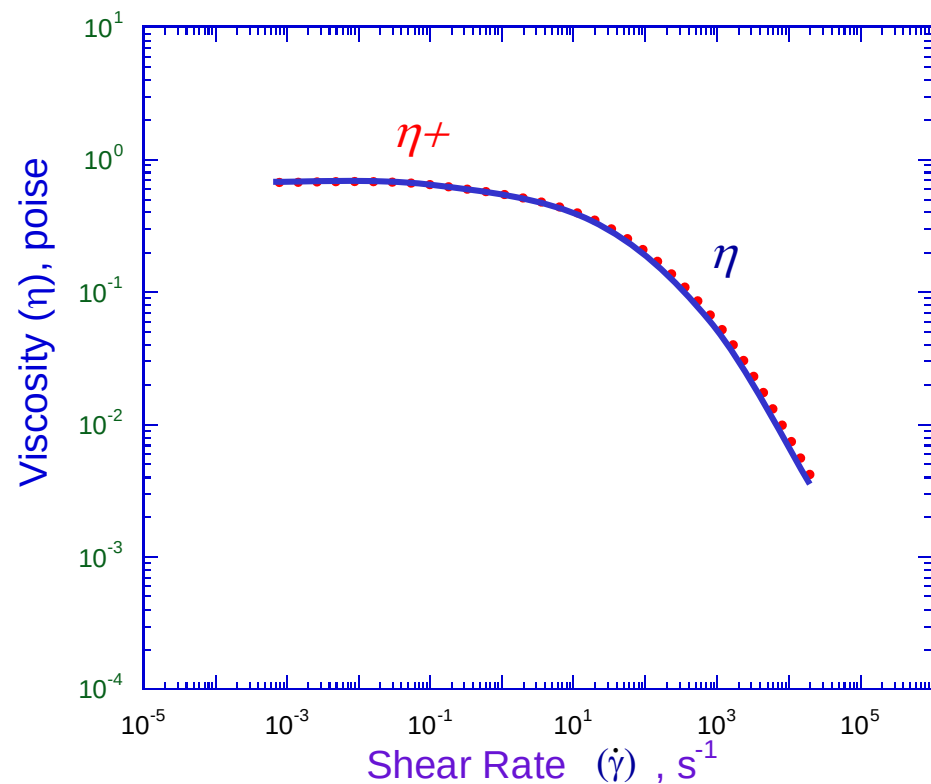
Start-up vs Steady shear

Transient (η^+) vs steady (η)

$$\eta(\dot{\gamma}) = \eta^+(t) \quad @ \quad \dot{\gamma} = 1/t$$

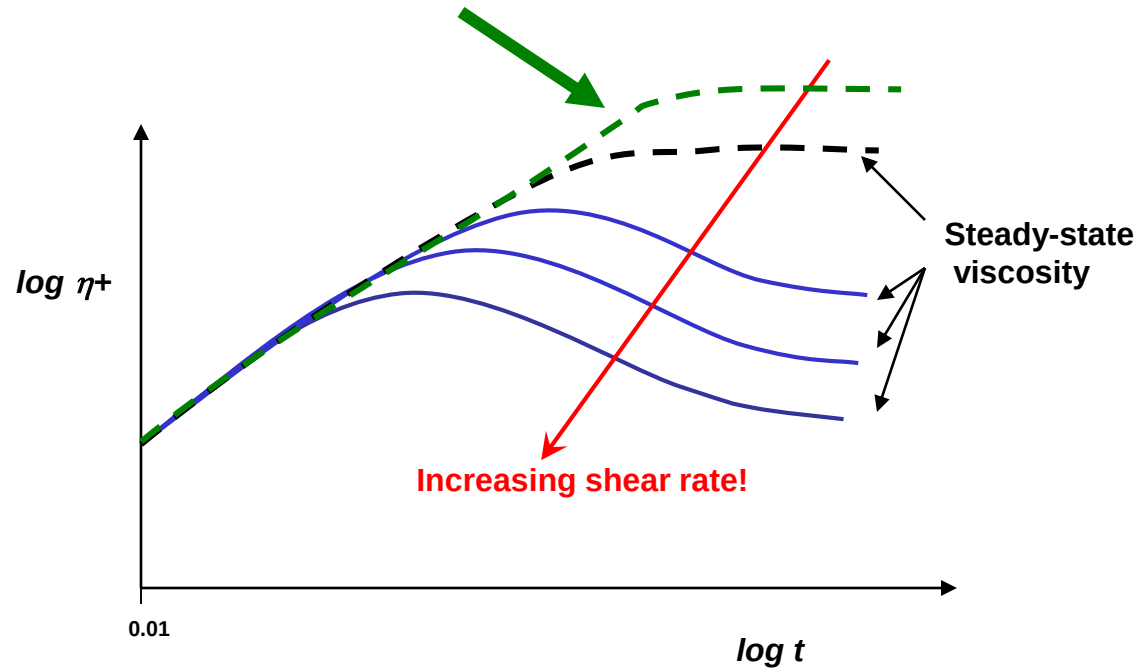
Gliese's Mirror Rule

η^+ is the limiting curve at low shear rates



Recall Stress Growth

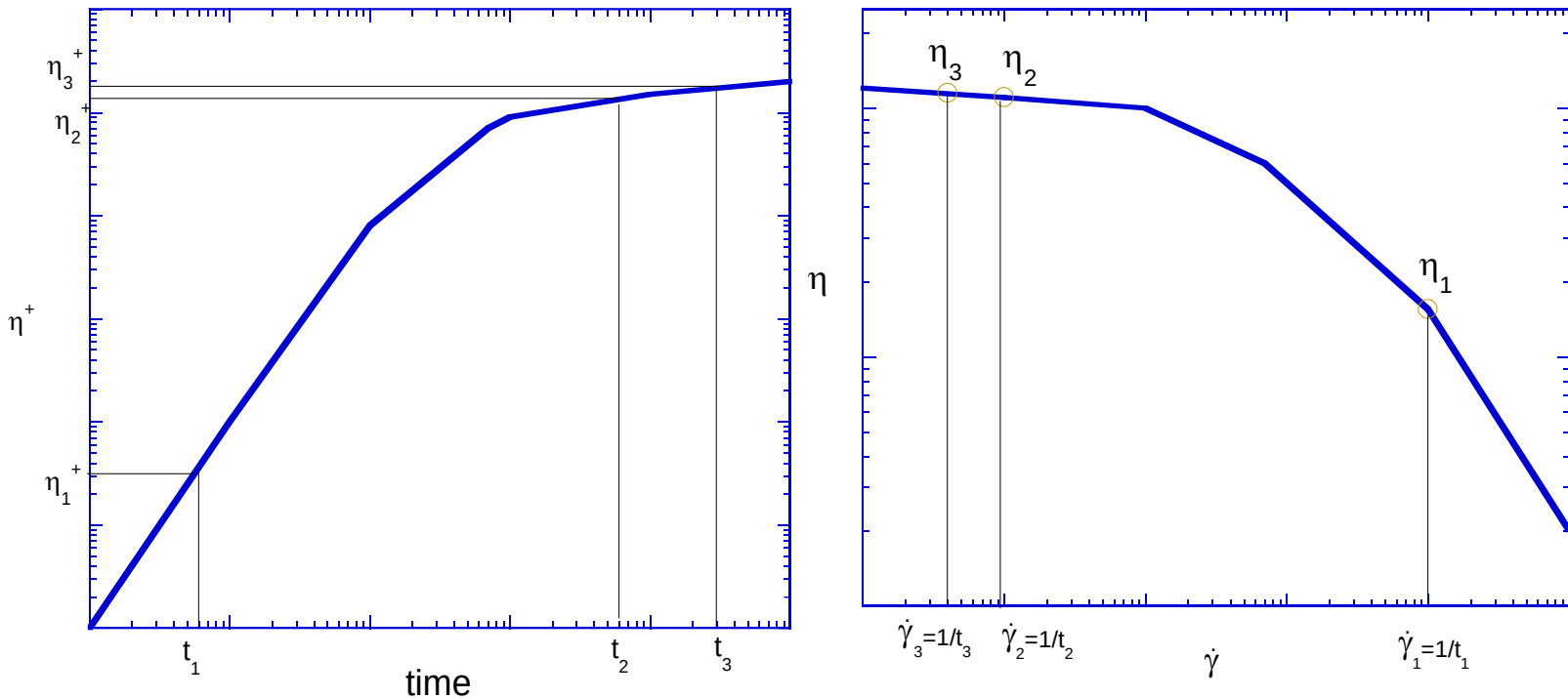
limiting curve at low shear rates



- Envelope corresponds to linear regime

Gliessele's Mirror rule

Relates stress growth data to steady shear viscosity



$$\eta^+(t) = \eta(\dot{\gamma}) \quad @ \quad \dot{\gamma} = 1/t$$

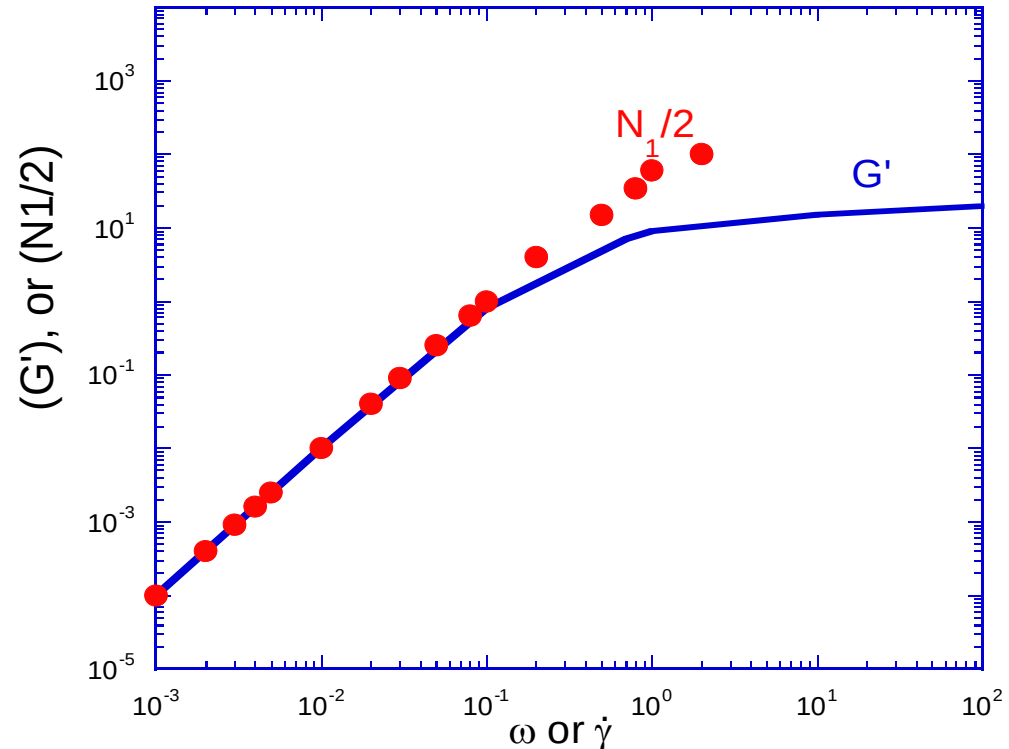
Normal Stress versus Elastic Modulus

- Normal stresses can have analogous behavior as G' :

$$N_1/2 = G'$$

- Good overlap at low frequencies or shear rate

- Deviation at high rates can be corrected using a modified eqn. (Laun, H.M.)



Note $N_1 = \tau_{xx} - \tau_{yy}$

Polymer Properties from Rheology

Dilute Solutions

$$[\eta] = K' M^a$$

*a = Mark Houwink exponent
has values between 0.5-0.8*

K related to polymer solvent interactions

Polymer Properties from Rheology

Molecular weight and Zero Shear Viscosity (Melt & Concn. Soln.)

Polymer Melt



$$\eta_0 \propto M_w$$

$$M < M_c$$

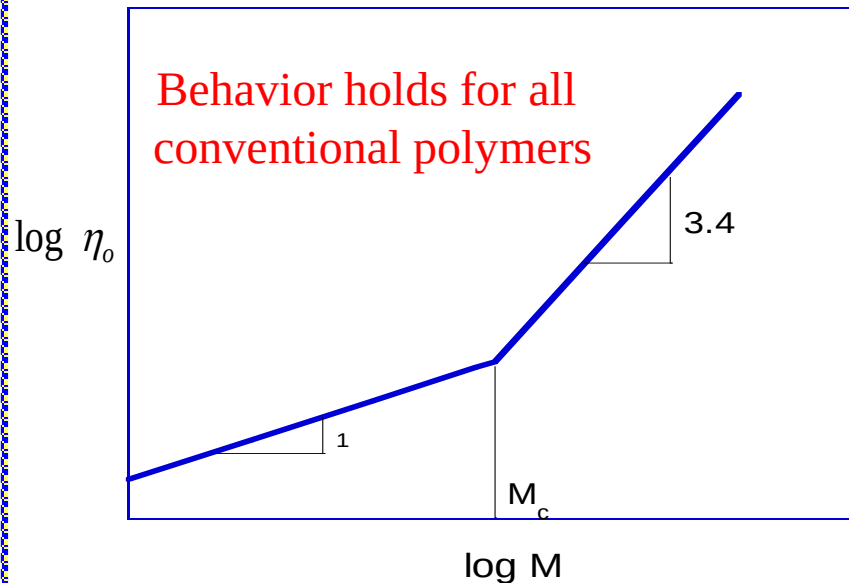
$$\eta_0 \propto M_w^{3.4}$$

$$M > M_c$$

Polymer Properties from Rheology

Molecular weight and Zero Shear Viscosity (Melt & Concn. Soln.)

Polymer Melt



$$\eta_0 \propto M_w$$

$$M < M_c$$

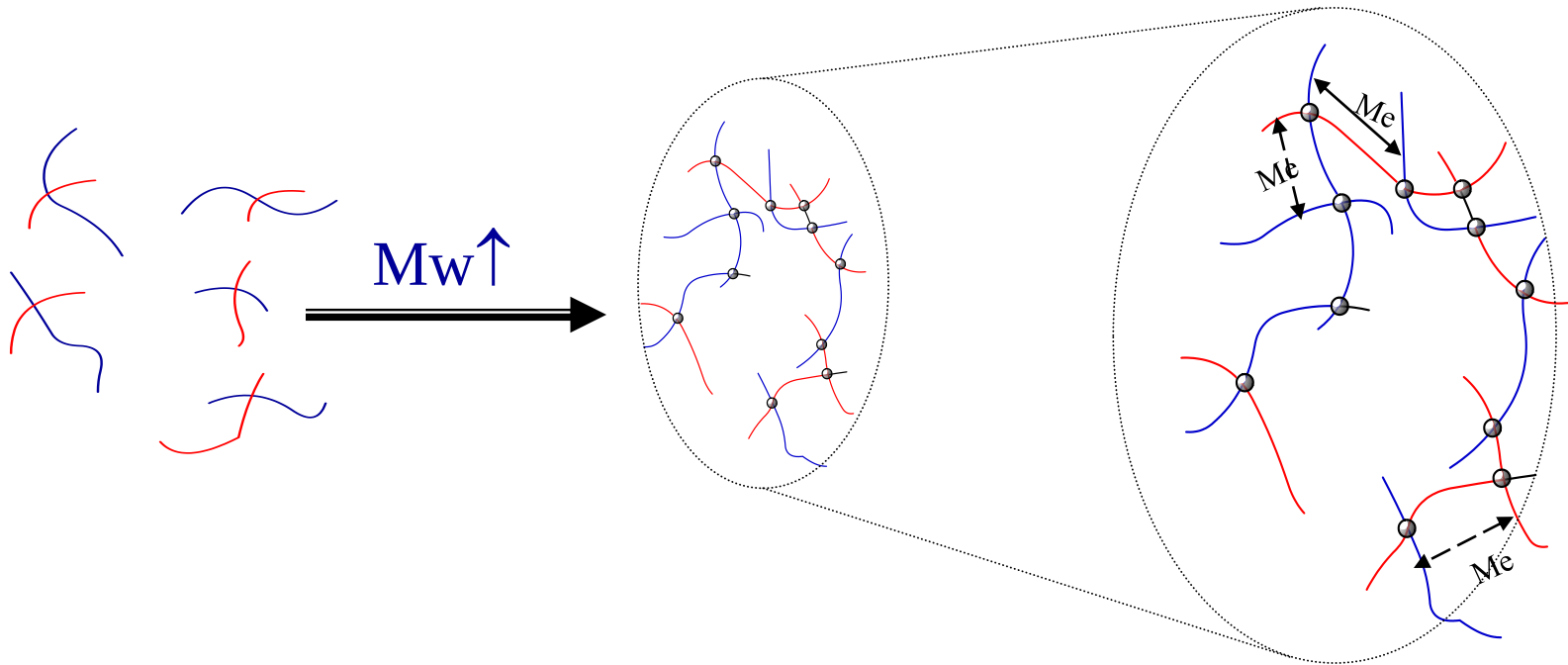
$$\eta_0 \propto M_w^{3.4}$$

$$M > M_c$$

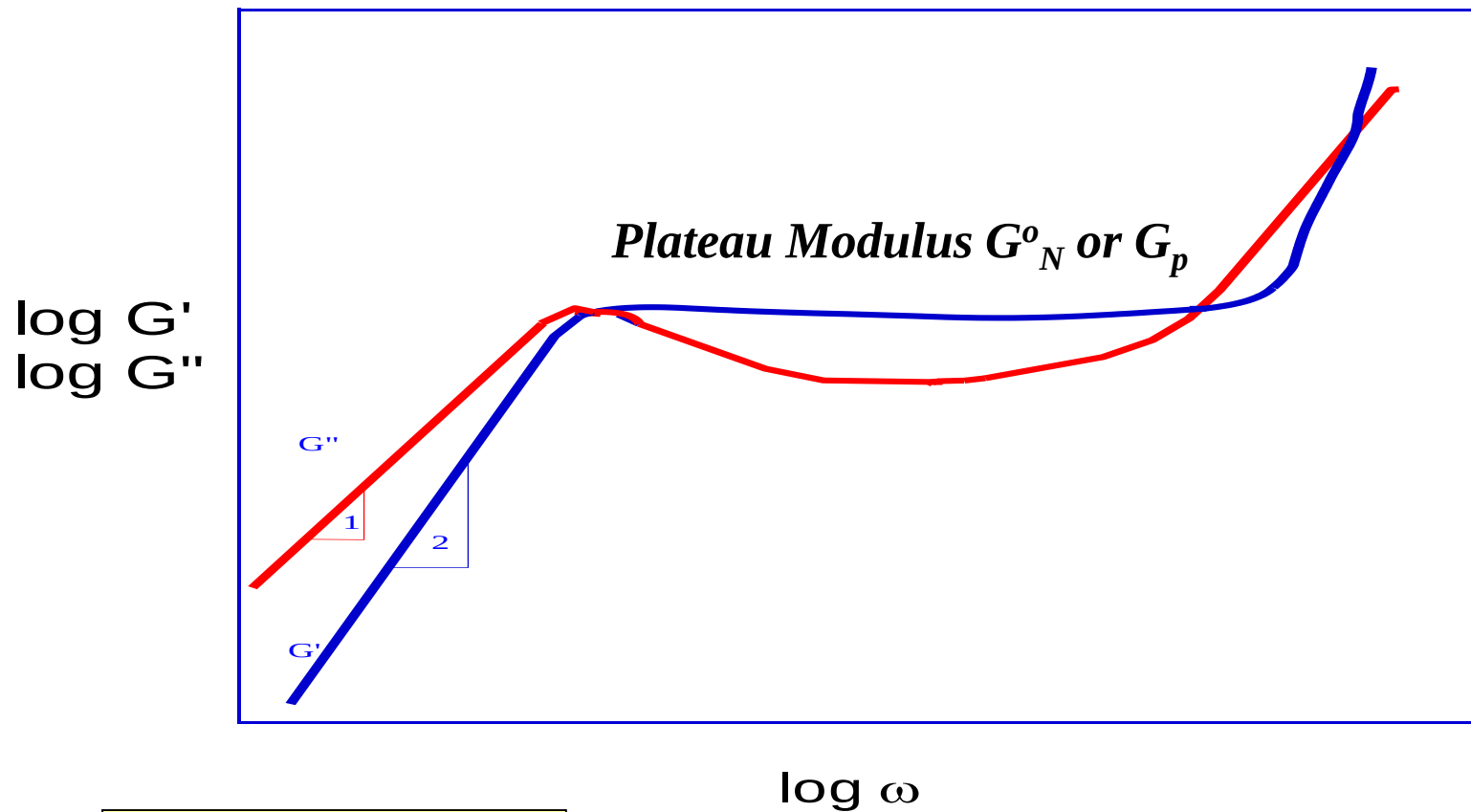
M_c Related to molecular weight between entanglements (M_e); $M_c = 2-3 M_e$

Critical and Entanglement Molecular Weights

$$M_c = 2-3 M_e$$

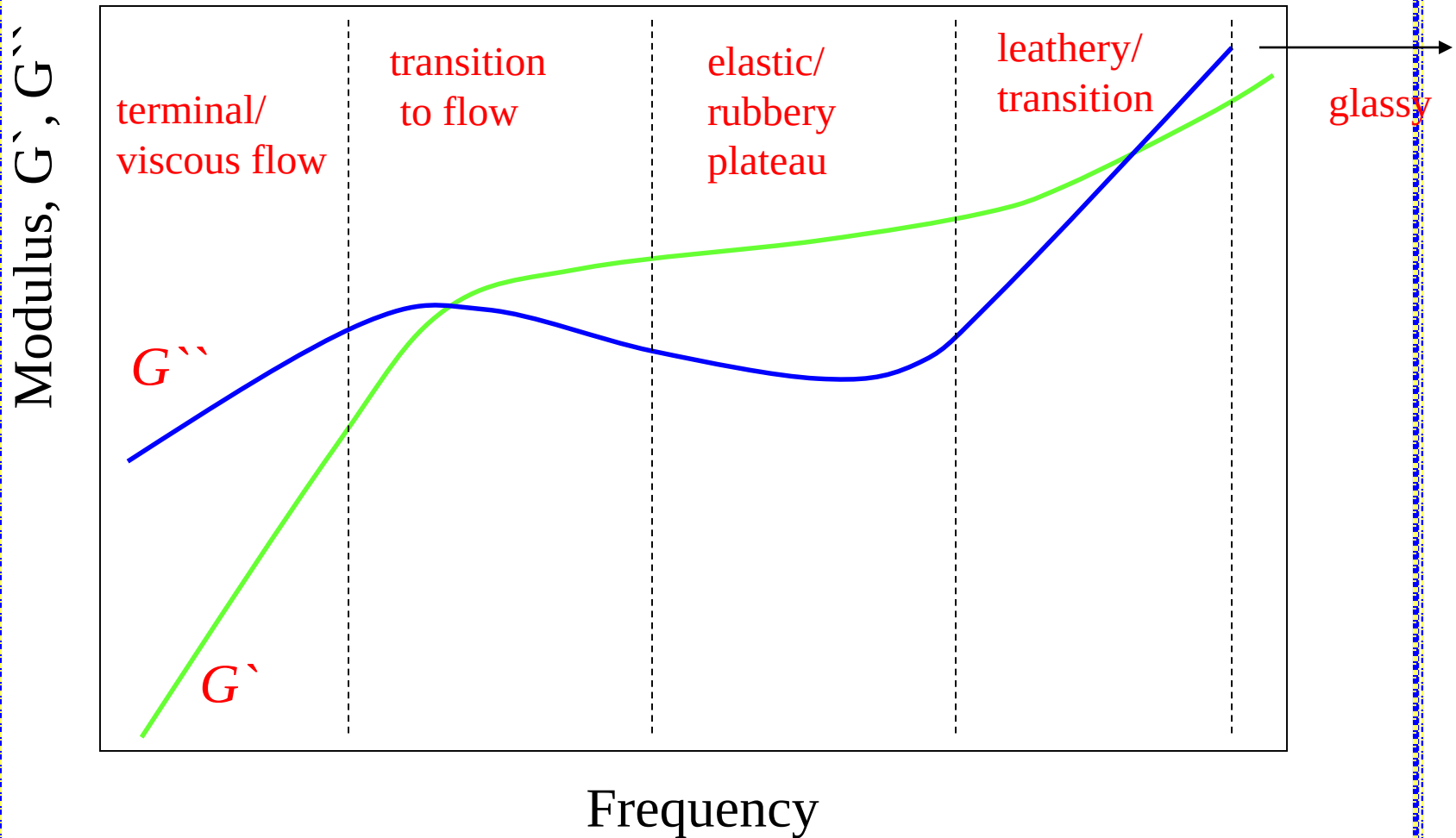


Elastic and Viscous Moduli (G' , G'')

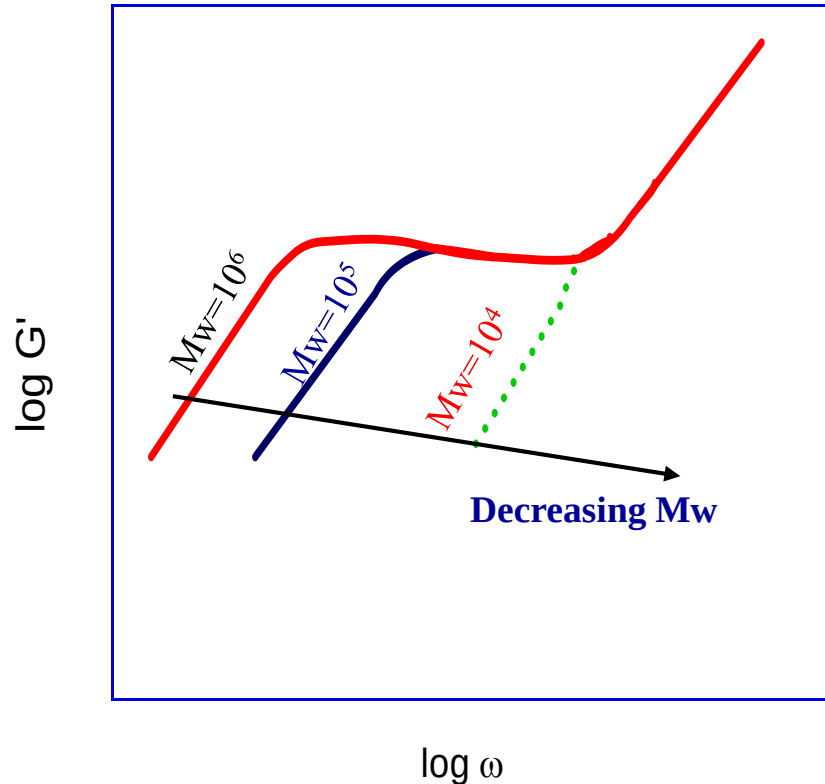


$$M_e \sim \frac{1}{G_N^0} = \frac{c \text{ (or } \rho) RT}{G_N^0}$$

Elastic and Viscous Moduli (G' , G'')



Elastic Modulus (G')



$$M_e \sim \frac{1}{G_N^0} = \frac{\rho RT}{G_N^0}$$

G_N^0 Plateau Modulus

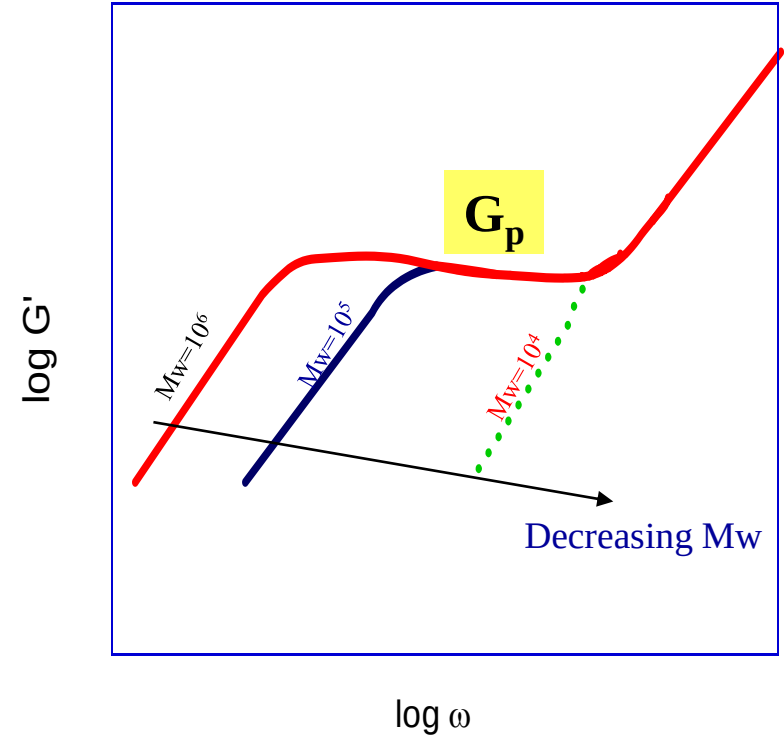
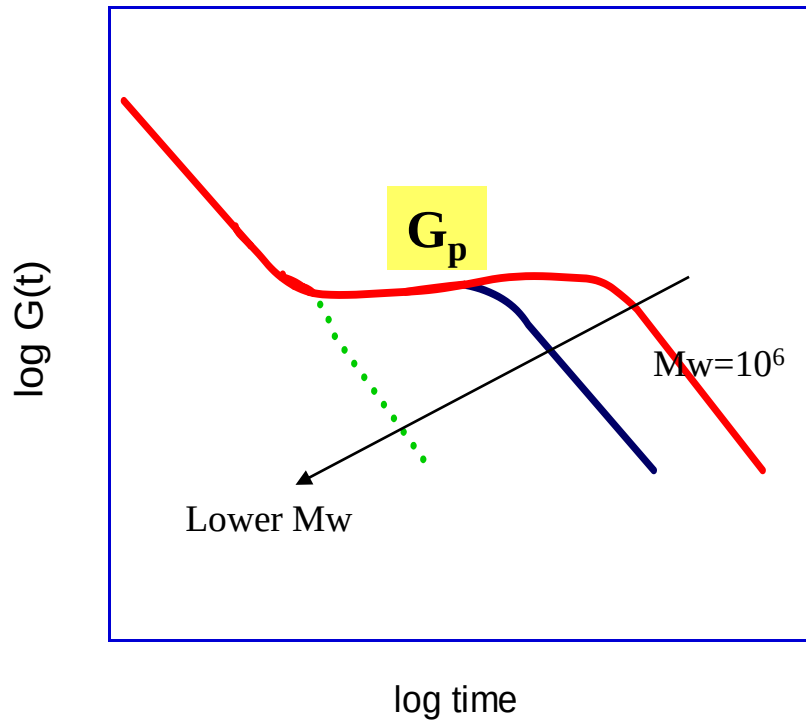
G_p or G_N^0

- related to entanglement mol. wt.
- independent of polymer M_w
- gets broader with increasing M_w .
- becomes less prominent with increased mol.wt. distribution

Terminal regime

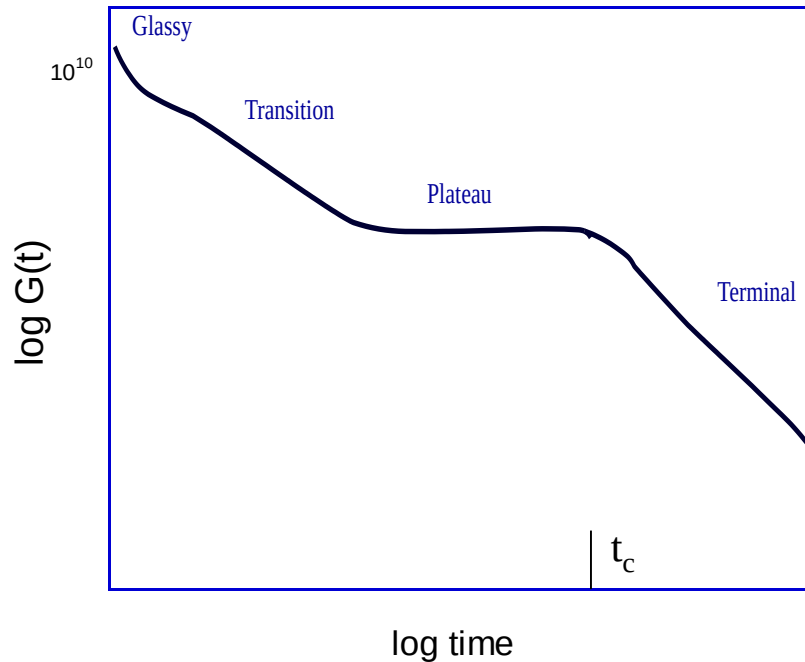
- shifts right with decreasing polymer M_w
- slope gets smaller than 2 with increasing polydispersity

Elastic and Relaxation moduli



- Same G_p for both
- Similar trends in both

Stress Relaxation (Step Strain)



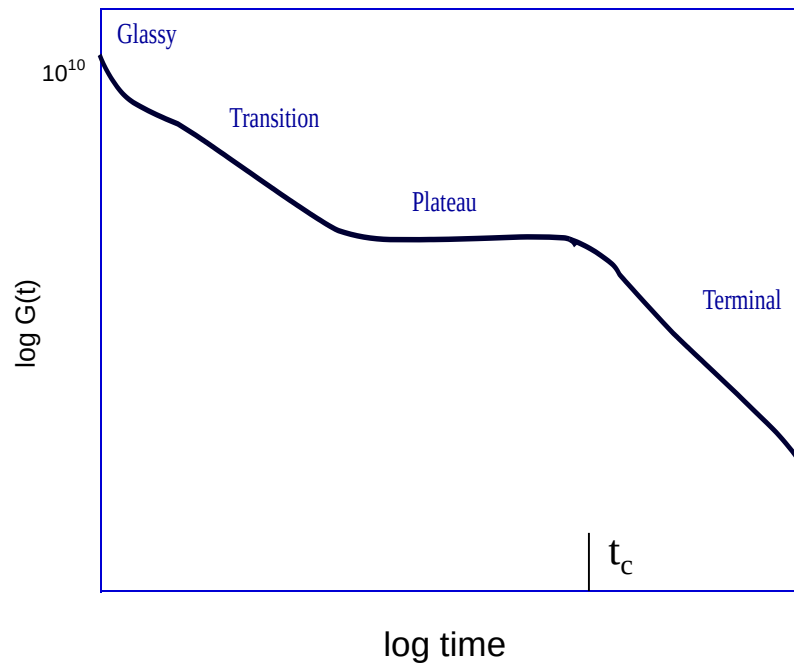
*Glassy- response glassy;
doesn't affect
flow; modulus $\sim 10^{10}$*

*Transition- short time
local relaxation*

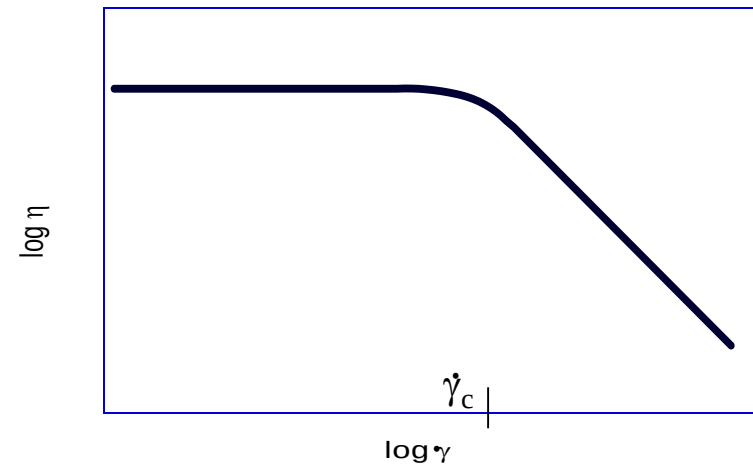
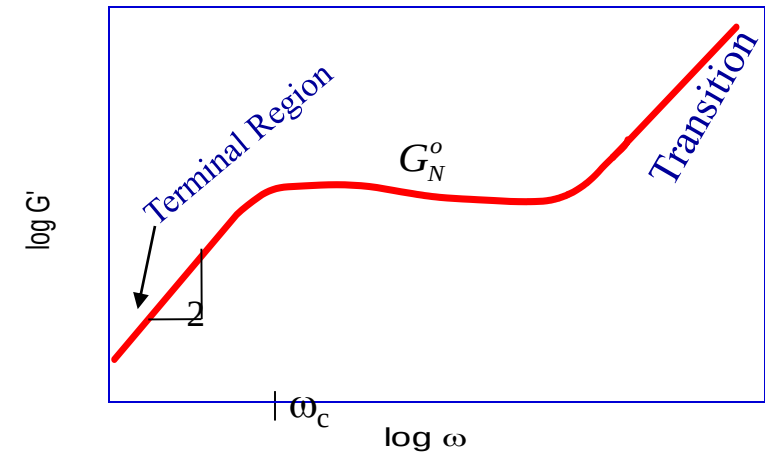
*terminal- long chain
relaxation; affects flow*

*plateau- separates short
time relaxation from long
time relaxation; elastic
behavior dominates*

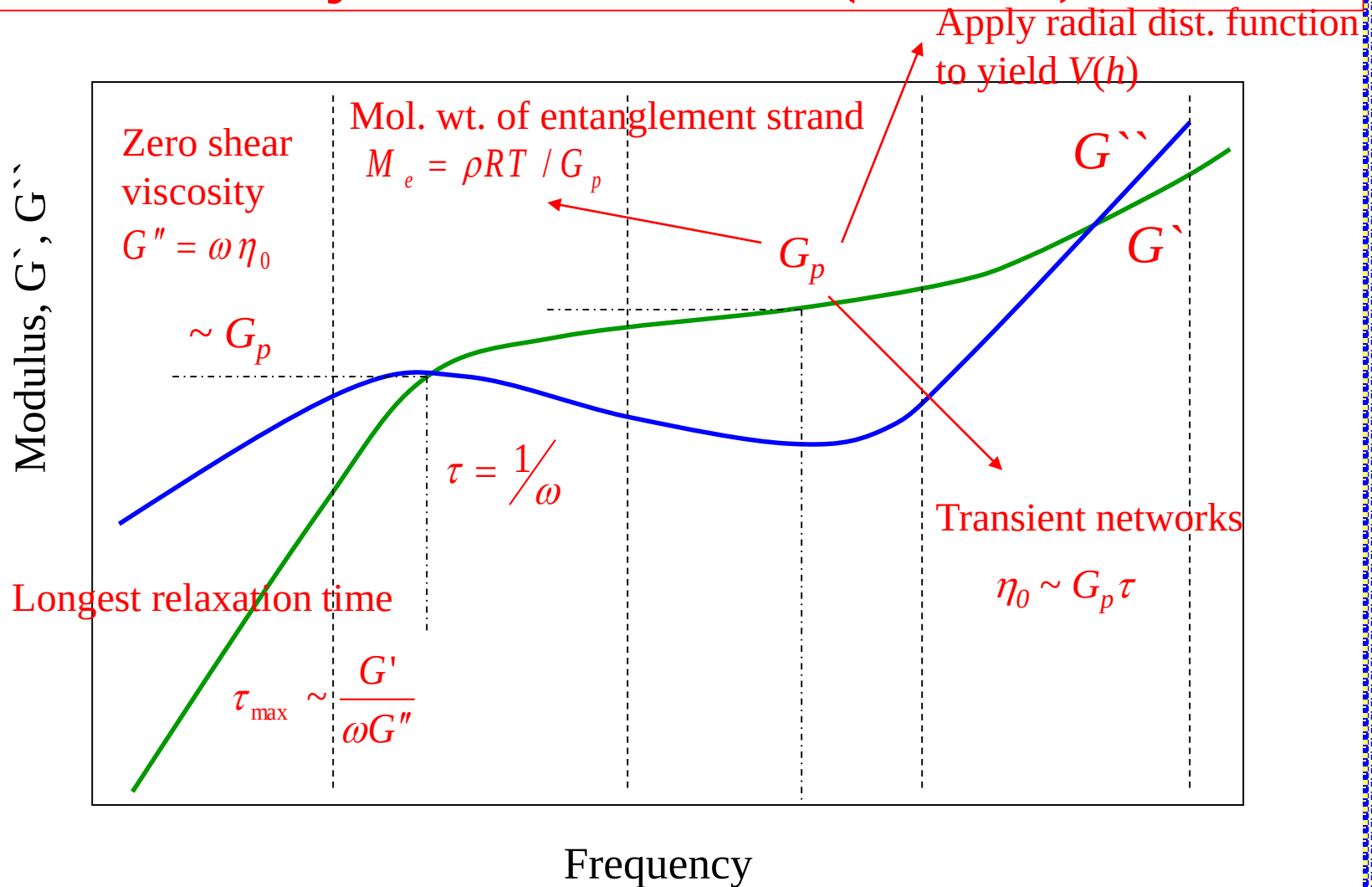
Correlation between experiments



$$\omega_c \cong \frac{1}{t_c} = \dot{\gamma}_c$$



Dynamic moduli (revisit)



Viscous or terminal region

- Here G'' predominates and energy is dissipated in viscous flow ($\delta \rightarrow 90^\circ$) and the stress has time to relax within the cycle of each oscillation.
- For very solid-like materials, the terminal region is very difficult to observe (long experimental times required).
- It is possible to estimate a value for the longest relaxation time of the system $\tau_{\max}$ from the value of $G'/(G''\omega)$.
- In the terminal region the frequency dependence of G'' is linear ($G'' \sim \omega$) and the frequency dependence of G' is quadratic ($G' \sim \omega^2$).
- The zero shear viscosity, η_0 , can be obtained as a low frequencies G'' is equal to $\omega\eta_0$

Transition to flow region

- This marks the end of the elastic dominated plateau region and is the region where G'' starts to become significant as the frequency is decreased (δ increases).
- The crossover frequency where $G' = G''$ ($\delta = 45^\circ$) is important as its reciprocal gives another measure of a network relaxation time for the system.
- In simple models of linear viscoelastic systems (e.g. Maxwell) this is the case.
- Similarly, G'' passes through a maximum in this regime and the reciprocal of the frequency associated with $G''_{\max}$ is taken by some workers as a network relaxation time.

Elastic/Rubbery Plateau Region

- Here elastic behaviour (energy storage) dominates and $G' > G''$ - within a given cycle of oscillation stress is supported by elements of the microstructure.
- G'' passes through a minimum and the frequency dependence of G' is weak.
- The characteristic value of G' in this region is the plateau modulus - G_p .
- G_p is a measure of the mechanical interaction density - e.g. from rubber elasticity theory $G_p = \nu_{eff} kT$ - ν_{eff} gives a measure of the number density of elastically active strands in the network. **(later)**
- For colloidal systems G_p is dependent on the radial distribution function of the particles and the force between them. **(later)**

Elastic/Rubbery Plateau Region

- Making a suitable approximation for the radial distribution function, it is possible to estimate the interparticle force. **(later)**
- Experimentally, G_p is estimated at the frequency at which G'' is minimum ($G''/G' = \tan \delta$ also passes through a minimum).
- Some workers measure G_p at the crossover frequency ($G' = G''$) as this is easier to access experimentally for polymer solutions with short relaxation times.
- For transient polymer networks - $\eta_0 \sim G_p \tau$ **(later)**

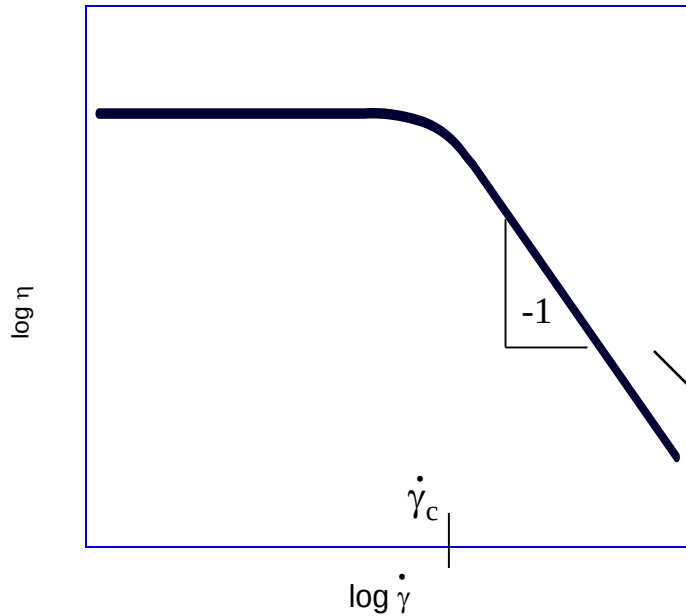
Transition and Glassy Regions

- At higher frequencies, viscous dissipation mechanisms again become more important and $G'' > G'$.
- There is a second crossover where $G' = G''$
- This region may be seen for polymer melts, but is seldom observed in the case of polymer solutions or aqueous complex fluids.
- At even higher frequencies the response of the system becomes glassy.

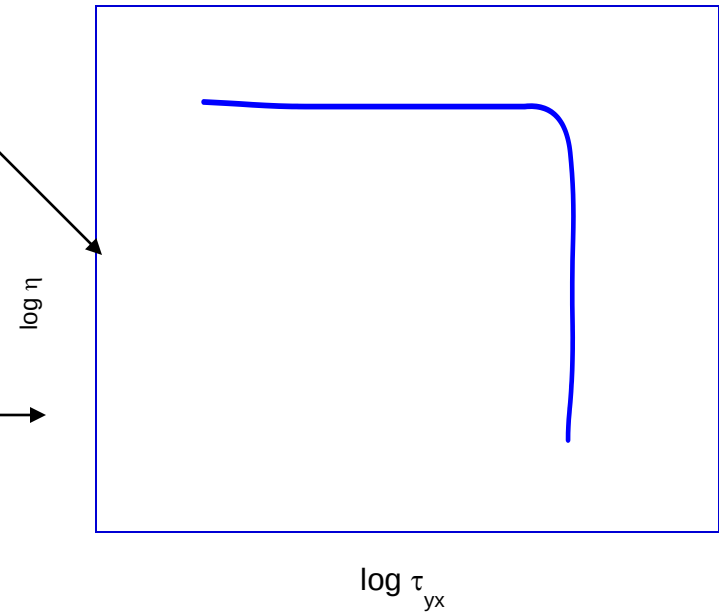
Extending available range of experimental frequencies

- Frequency range of modern rotational instruments $10^2 \rightarrow 10^{-3} \text{ rad s}^{-1}$
- For most materials only ~ 2 of the different characteristic “zones” of behaviour are seen in a given experiment (particularly if only 2 or 3 decades of ω are employed)
 - - e.g. terminal and transition to flow for a polymer solution ($\omega\tau_{max} \sim 1$), plateau and transition for a polymer melt ($\omega\tau_{max} \gg 1$).
- *Methods of reduced variables* may be used to extent ω range.
- Assumes all relaxation processes have similar dependence on T or c (*time-conc. or time-temperature superposition*).
- Superpose data from experiments at different T or c to give extended range of experimental frequencies.

Exercise



Need to plot η versus stress (τ_{yx})



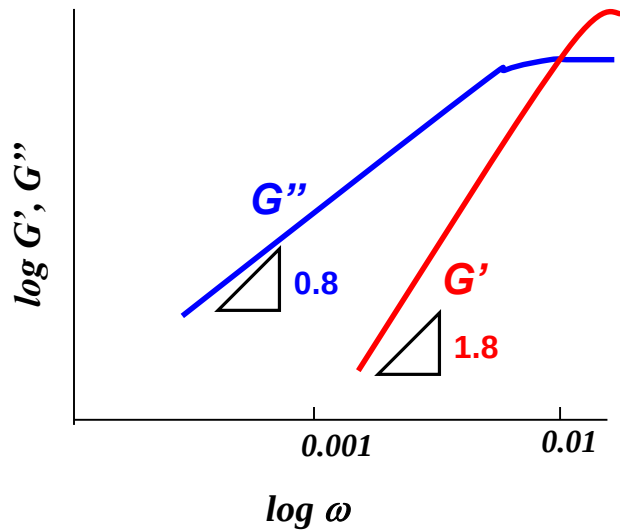
at the -1 slope area

$$\eta = \text{constant} * \dot{\gamma}^{-1}$$

$$\tau_{yx} = \eta \dot{\gamma} = \text{constant}$$

In-class problem

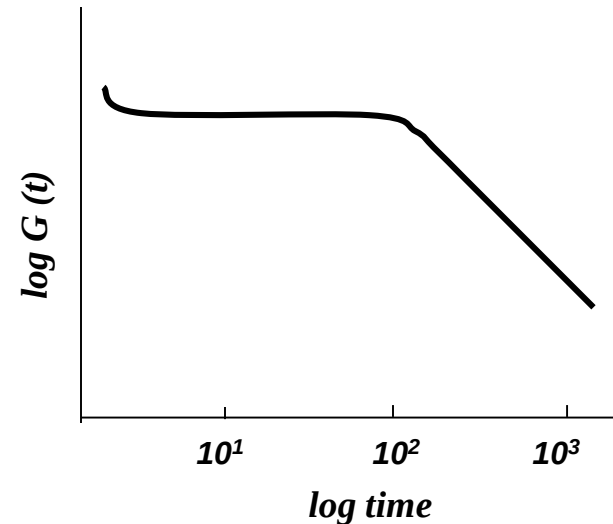
- New mat'l send to 2 labs. Results shown from the 2 labs.
- What information can you back out about the polymer?



Relaxation time ~ 100 sec

Viscosity – Cox Merz rule

Sample has polydispersity



Entanglement Mol. Wt.- from G_p

Normal stress: $N_1/2 \approx G'$ for some range

Last Class & Today- Shear experiments

- Correlation between Material functions
- Effects of molecular properties on Material functions

Problem 1

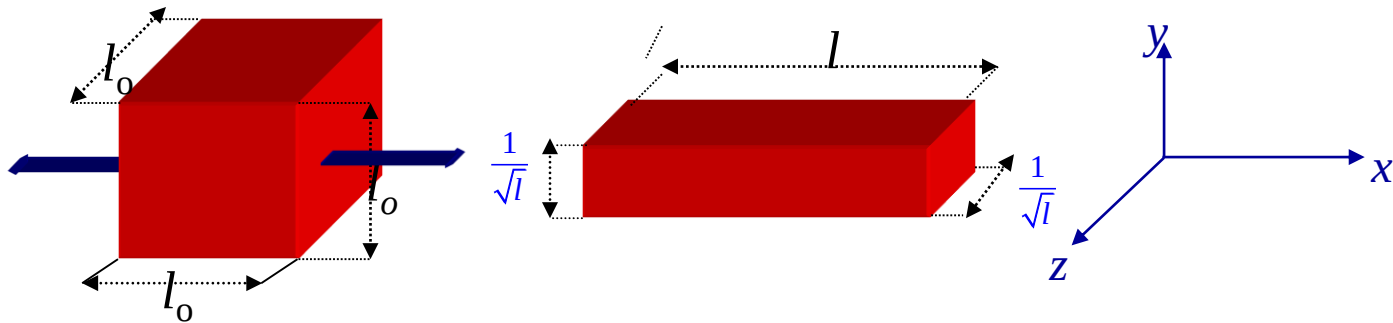
$$\underline{\underline{T}} = \begin{bmatrix} -7 & 5 & 0 \\ 5 & -7 & 0 \\ 0 & 0 & -7 \end{bmatrix}$$

*Given that we have a shear flow,
comment on the nature of the fluid*

$$\underline{\underline{T}} = \begin{bmatrix} 20 & 5 & 0 \\ 5 & 19 & 0 \\ 0 & 0 & 19 \end{bmatrix}$$

*Given only the stress tensor info,
Comment on flow type and fluid*

Uniaxial Elongational Flow



$$\tau_{xx} - \tau_{yy} = \bar{\eta}(\dot{\epsilon})\dot{\epsilon},$$

$\dot{\epsilon}$ is the extension rate

$$v_x = \dot{\epsilon}l$$

$$v_y = -\frac{1}{2} \dot{\epsilon}l$$

$$v_z = -\frac{1}{2} \dot{\epsilon}l$$

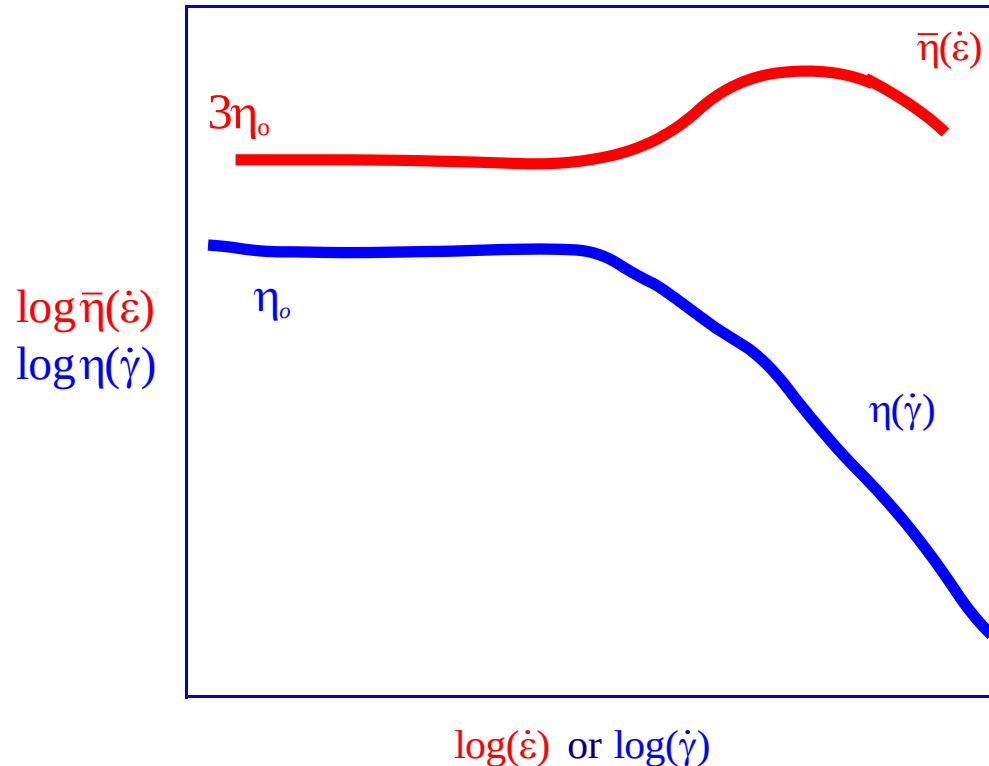
Recall for Shear

$$\tau_{yx} = \eta(\dot{\gamma})\dot{\gamma},$$

$$v_x = \dot{\gamma}y \quad (v_y = v_z = 0)$$

How does $\bar{\eta}$ change as function of $\dot{\epsilon}$?

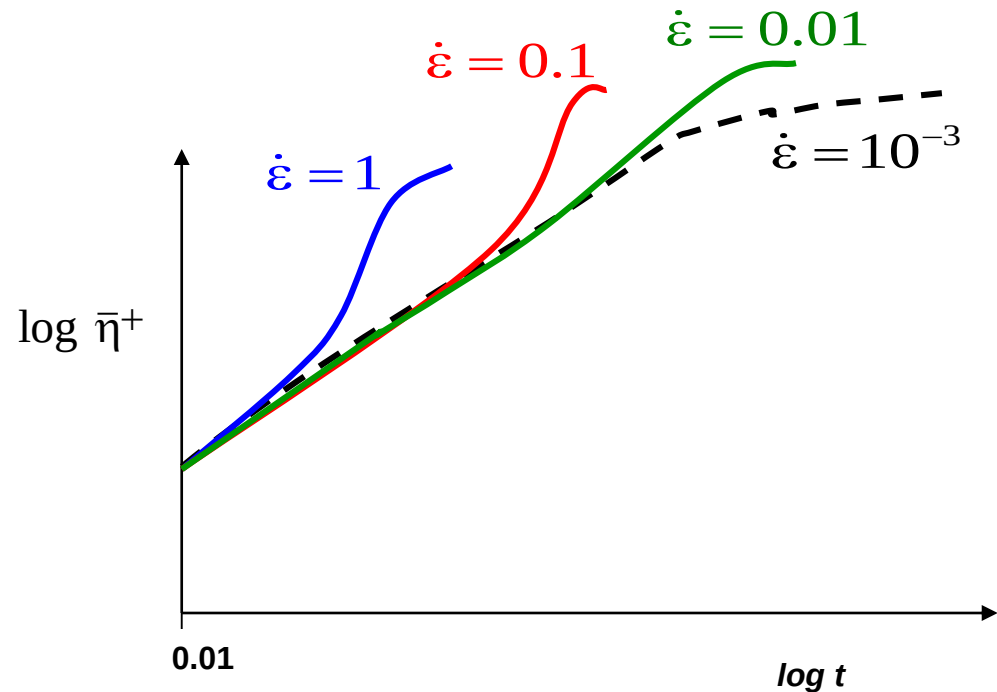
Extensional vs Shear viscosity



Trouton's rule *at* low $\dot{\epsilon}$ and $\dot{\gamma}$; $\bar{\eta}(\dot{\epsilon}) = 3\eta(\dot{\gamma})$

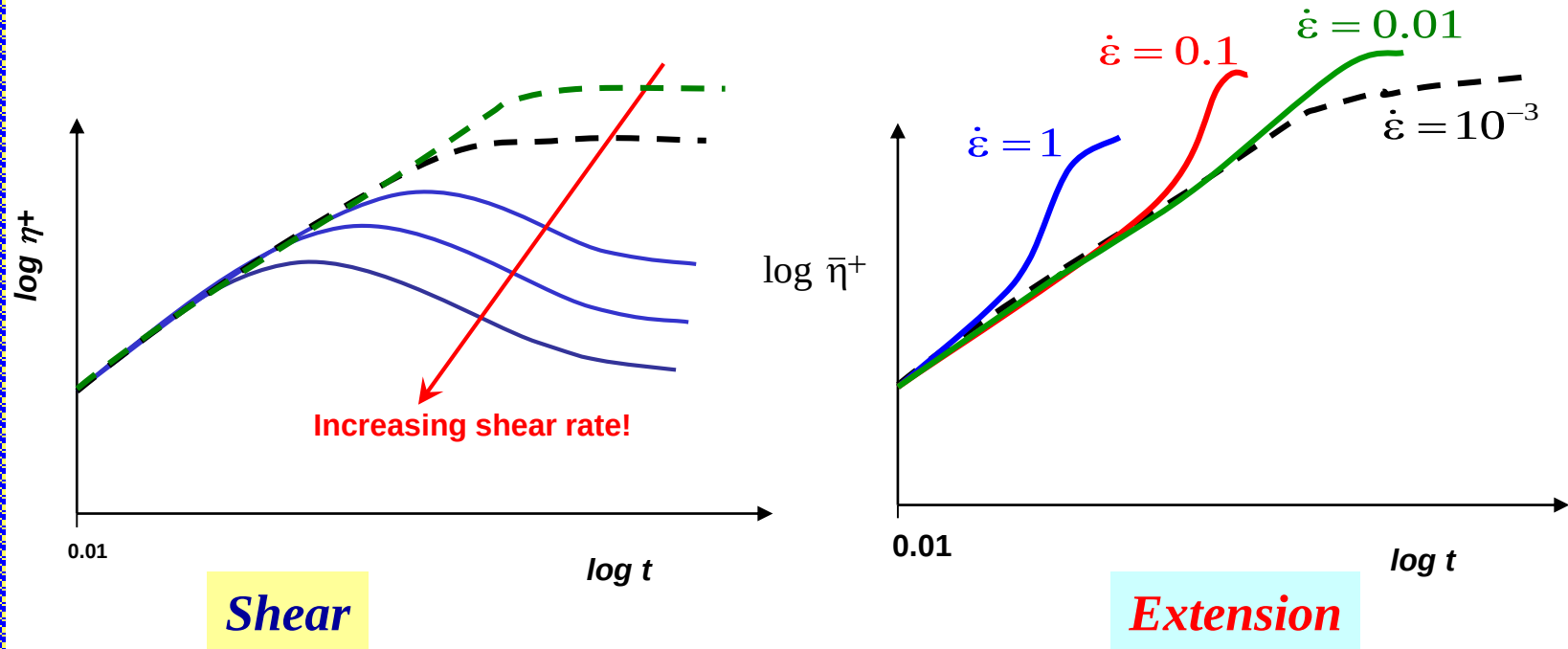
Transient extensional viscosity

- Low strain rate corresponds to linear regime
- Viscosity shows up turn with increasing strain rate
- Upturn is called strain hardening
- Strain hardening occurs at same Hencky strain



• Hencky strain $\epsilon = \dot{\epsilon} t$

Transients: Shear vs Extension



- Strain hardening (extension) vs softening (shear)
- Envelope corresponds to linear regime
- How are η^+ and $\bar{\eta}^+$ related at low strain rates?

Summary & Moving Forward

- Materials are viscoelastic: Fading memory, elastic recoil, rod climbing, shear thinning
- Examined material functions: Viscosity, Normal stress difference, G' , G'' , etc



- Can we use models to predict shear thinning? Macosko, Ch 2
- Can we use models to mimic viscoelasticity? Macosko, Ch 3

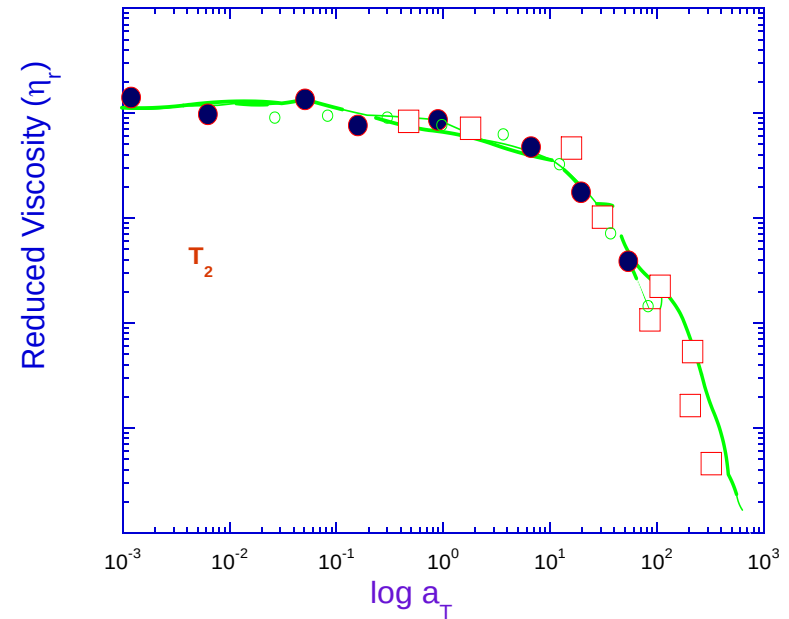
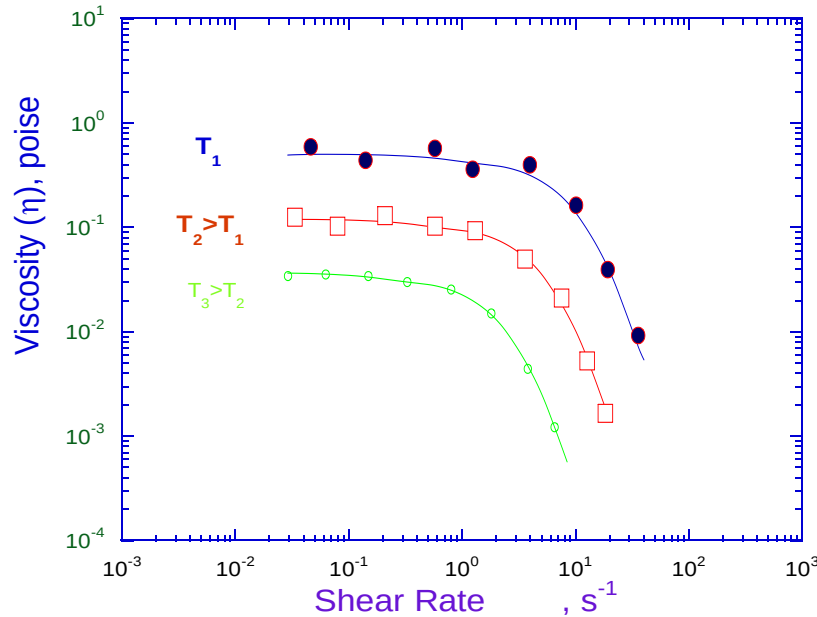


- How do we measure material properties? Rheometry-Macosko Ch 5-7



- Interesting materials & phenomena: gels, thixotropy, colloids,....

Time-Temperature Superposition



$a_T \equiv$ Shift factor

$$a_T = \frac{\eta_o(T)}{\eta_o(T_{ref})} \frac{T_{ref}}{T}$$

$$\begin{aligned} \eta_r &= \eta(\dot{\gamma}, T) \frac{\eta_o(T_{ref})}{\eta_o(T)} \\ &= \eta(\dot{\gamma}, T) \frac{T_{ref}}{T a_T} \end{aligned}$$

Shift Factor

How does a_T changes with temperature?

$$a_T = \exp \left[\frac{\Delta H}{R} \left(\frac{1}{T} - \frac{1}{T_{ref}} \right) \right]$$

$\Delta H \sim$ Activation Energy of flow

Typically for low Mw materials and polymer 100°K
above T_g (*glass transition temperature*)

$$\log a_T = \frac{-C_1^o (T - T_{ref})}{C_2^o + (T - T_{ref})} \quad \text{WLF equation}$$

Valid when sample temperature is between T_g and $T_g + 100$